Vampire Paper Post-Mortem

(and next steps on my PhD)

Introduction

The final paper was submitted to the Vampire workshop! This is the first big win of my PhD, and the process of getting there allowed me to improve the core objectives that will be useful for writing the first-year report.

This one-pager presents some thoughts on the entire process to use as a guide for the report.

Timeline

Michaelmas

— Went over bibliography related to connection provers.
— Forked Fredrik's repo, and started experimenting with it.
— Created the ClassicalSAT calculus on this fork, based on SATCoP[1].

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— New, major fork in the connections prover (`MyPyCoP`).
— FOF translation in `MyPyCoP`.
— Figured out how to properly work the HPC. Created runner scripts to easily batch many experiments simultaneously.
— Strategies in `MyPyCoP`, and the `compare_strategies.py` script.
— These two changes made it possible to test and iterate faster.

Easter Vacation + Easter

— Began writing weekly A4 papers during weekly meetings. Good idea!
— General experimentation in different areas.
1. Different schedulers, including using ParamILS[2].
2. PySATCoP and PySATCoPLiteral, strategy-level implementations of SATCoP.
3. Different cut-based policies: LeanCoP-style cuts, reduction-only cuts, ...
4. Different reset policies.
5. One-armed-bandit and reset-based policies.
6. SAT-based features for tableau guidance.

— **April 29**: first version of the strategy that later became SATResetCoP. Publishing this to FLoC became my main priority.
— Ran experiments on current environment to ensure the new strategy works correctly and there was no "simpler" competitor.
— **May 8 – 28** Wrote Vampire paper.
— Worked with Fredrik to improve the paper and make it closer to the standard. Note that this is the first proper publication of the connections prover[3]!

Thoughts and Questions

- Should I have spent less time experimenting and setting up the environment, and more time researching? I feel that "wide" paper discovery became a lot slower after Michaelmas, but I'm not sure how much I could have gained after reading the core papers.
- Did I get too focused on plots and tables? Maybe I should have done this at the end.

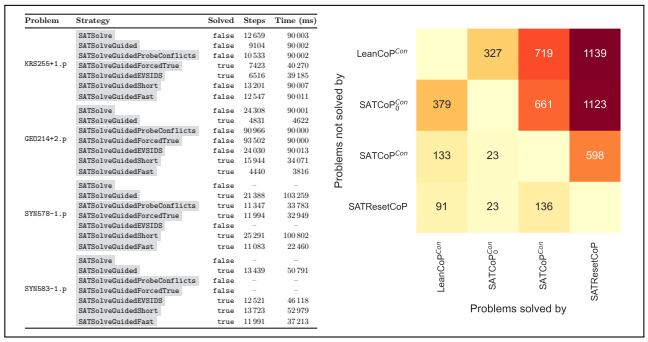


Figure 1: Example of the tables and plots I spent time producing. Were these the right level of analysis, or should I have left this for writing the paper?

- I should definitely have started working with Fredrik earlier. Big review last week before submission; he could also have helped me during experimentation.
- Focusing on building strong testing infrastructure *first* was a great idea. Experimenting quickly is always good.
- I'm not sure how I could have made it on time for the UNIF workshop. Would it make sense to keep a few experiments "in reserve" waiting for these deadlines?

Next Steps

1. Current priority: **First-year Report**. Deadline June 30th, so not a lot of time to go.
2. Let's send the current tableau prover to CASC! I'll start the work on this soon.
3. How different does this paper have to be to send these experiments to another conference or journal?

Tiny References

[1] Michael Rawson and Giles Reger. "Eliminating Models During Model Elimination". In: *Automated Reasoning with Analytic Tableaux and Related Methods*. Ed. by Anupam Das and Sara Negri. Vol. 12842. Cham: Springer International Publishing, 2021, pp. 250–265. ISBN: 978-3-030-86058-5 978-3-030-86059-2.

[2] Frank Hutter et al. "ParamILS: An Automatic Algorithm Configuration Framework". In: *Journal of Artificial Intelligence Research* 36 (Oct. 30, 2009), pp. 267–306. ISSN: 1076-9757. arXiv: 1401.3492 [cs].

[3] Fredrik Rømming, Jens Otten, and Sean B. Holden. "Connections: Markov Decision Processes for Classical, Intuitionistic and Modal Connection Calculi". In: *Proceedings of the 1st International Workshop on Automated Reasoning with Connection Calculi*. Vol. 3613. CEUR Workshop Proceedings. 2023, pp. 107–118.